Reference: Collaborative Project/2022 **Date:** August 28, 2024

RESEARCH PROJECT COMPLETION CERTIFICATE

This certifies that the collaborative research project titled "**Development of a Deep Learning-Based Eye Diagram Analyzer for OptiSystem Software**" has been successfully completed under the expert leadership of:

- **Principal Investigator:**
  - **Dr. Aadil Raza**, Department of Physics, COMSATS University Islamabad (CUI), Islamabad, Pakistan
- **Co-Principal Investigators:**
  - **Dr. Basit Raza**, Department of Computer Science, COMSATS University Islamabad (CUI), Islamabad, Pakistan
  - **Dr. Muhammad Iqbal**, Department of Computer Science, National University of Technology (NUTECH), Islamabad, Pakistan
  - **Dr. Saeed Iqbal**, Department of Computer Science, Barani Institute of Information Technology, Rawalpindi, Pakistan

The project began on August 23, 2022, and concluded on August 22, 2024, spanning a duration of 2 years. Throughout the project, the development and integration of a deep learning-based eye diagram analyzer for OptiSystem Software were successfully achieved. The research team also published a journal paper (*A. S. Al-Shamayleh, A. Raza*, Z. Ali, S. Malik, S. Iqbal, B. Raza, M. Iqbal, and A. Atieh "Performance Evaluation of Optical Transmission Based on Link Estimation by Using Deep Learning Techniques", IEEE Access, vol. 12, pp. 64126-64139, 2024*) which details the key findings and results of the project.

We extend our congratulations to the entire team for their outstanding performance on this innovative research project.

On behalf of Optiwave System Inc.

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